TEXORA · LECTURE SHEET

Process Control in Draw Frame

Draw frame control has two distinct jobs: hold the delivered sliver weight exactly on target (the A% test and draft correction), and keep drafting faults — waves and periodic defects readable on a spectrogram — out of the sliver. Both are covered here with the standard calculations.

YM-4101 · PROCESS CONTROL IN SPINNING · TOPIC 5 · SPECIALIZATION — YARN MANUFACTURING

1. Weight control: the A% test

Sliver weight is tested by cutting and weighing a standard length (usually 5 m, or 1 m repeated) from each delivery on a schedule. The deviation from nominal is expressed as A%:

$$A\% = (\text{measured weight} - \text{nominal weight}) / \text{nominal weight} \times 100$$

WORKED EXAMPLE — A% AND THE DRAFT CORRECTION

Nominal finisher sliver = 3.90 g/m. The shift test gives 3.78 g/m. Total draft in use = 8.2.

A% = (3.78 − 3.90)/3.90 × 100 = **−3.08%** (the sliver is light).

Delivered weight is inversely proportional to draft, so to raise the weight by the ratio 3.90/3.78 the draft must fall by the same ratio:

New draft = 8.2 × (3.78/3.90) = **7.95**.

The correction is made on the draft change wheel (or keyed into the machine control), and — the step most often skipped — a fresh test is taken after the change to confirm the sliver is back inside the action band, typically ±1% of nominal. Note that a −3% sliver left uncorrected becomes a −3% yarn count error at the end of the line: drawframe weight control *is* count control.

Before correcting, check the feed: if card sliver has itself drifted light, correcting at the drawframe restores the number but hides an upstream fault. The control discipline is to test feed and delivery together and correct at the process that drifted.

2. The spectrogram — reading drafting faults

The evenness tester draws a spectrogram: the amplitude of mass variation against wavelength. Mechanical faults appear as sharp peaks ("chimneys") at wavelengths fixed by the rotating part that

causes them; drafting faults appear as broad humps.

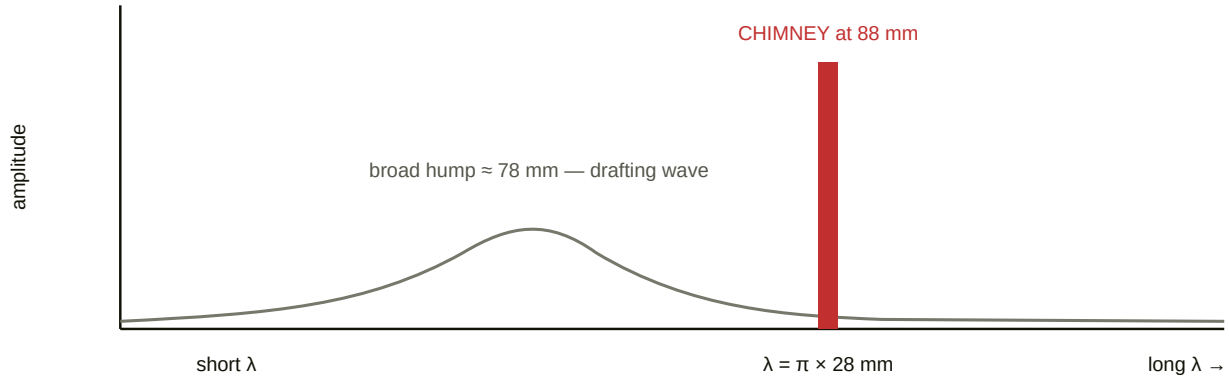


Figure 1 — Stylised spectrogram of a drawframe sliver. The broad hump near 78 mm is the drafting wave; the sharp chimney at 88 mm is the signature of the 28 mm front roller (λ = πD).

WORKED EXAMPLE — LOCATING A PERIODIC FAULT

A spectrogram of finisher sliver shows a chimney at wavelength 88 mm, and a broad hump centred near 78 mm.

Periodic fault: λ = π × D → D = 88/π = **28 mm** — exactly the front bottom roller diameter. The roller is eccentric, bent or carrying a damaged cot; it repeats its fault once per revolution, and the sliver carries one fault every 88 mm. Action: replace or true the roller/cot; the chimney should vanish on retest.

The hump at 78 mm ≈ 2.8 × mean fibre length (28 mm) is the classic **drafting wave** — floating fibres surging between roller pairs. Action: check roller settings (too wide for the staple), top-roller pressure and break draft.

3. Routine controls and norms

CONTROL	METHOD	TYPICAL NORM / ACTION
Sliver weight (A%)	Cut-and-weigh each delivery, per shift	Within ±1%; correct draft as worked above
Sliver U% / CV	Evenness tester, per machine per week (or online)	Finisher sliver U% ≲ 3%; investigate rising trend
Spectrogram	With each evenness test	No chimneys; humps explained by fibre length
Roller condition	Cot inspection, eccentricity check	Replace cots on schedule, not on failure

Stop motions & autoleveller	Function test; leveller response test	Must trip on sliver break and on empty can

Roller settings are verified against fibre length whenever the mixing changes staple: the main-zone setting should exceed the mean fibre length by the standard allowance for the roller system; a setting carried over from a longer-staple mixing is a drafting-wave generator.

4. Breaker versus finisher duties

The breaker passage takes card sliver with its hooks leading in the unfavourable direction and does the heavy doubling work; the finisher, often autolevelled, owns the final weight accuracy — which is why the A% regime is strictest there. A fault split: weight drift appearing in finisher delivery but not breaker delivery localises the investigation to one machine; drift in both points back to the card room (Topic 4).

Glossary

A% — the percentage deviation of tested sliver weight from nominal. **Spectrogram** — the plot of mass-variation amplitude against wavelength from an evenness tester. **Chimney** — a sharp spectrogram peak caused by a periodic mechanical fault. **Drafting wave** — the broad spectrogram hump from uncontrolled floating fibres, centred near 2.5–3 × mean fibre length. **Cot** — the synthetic rubber covering of a top drafting roller. **Eccentricity** — a roller rotating about an axis offset from its geometric centre, printing a fault every revolution. **Autoleveller** — the device trimming draft continuously to hold delivered weight.

Key points

• A% = (measured − nominal)/nominal × 100; sliver at −3.08% is corrected by draft 8.2 → 7.95, then re-tested.
• Correct at the process that drifted — test feed and delivery together before touching the draft.
• Periodic faults read off the spectrogram: $\lambda = \pi D$ (88 mm → the 28 mm front roller).
• The drafting wave hump sits near 2.5–3 × fibre length; its growth means settings, pressure or break draft need attention.
• The finisher passage owns final weight accuracy; its ±1% band is the mill's first count-control gate.

Practice

1. Nominal sliver is 4.00 g/m; the test gives 4.06 g/m with draft 7.6 in use. Calculate A% and the corrected draft.

2. A spectrogram chimney appears at 110 mm in roving-frame feed sliver. Which roller diameter does it indicate, and what defect do you expect to find?
3. Why must a post-correction retest be taken from a fresh length of sliver rather than the material already in the can?

Answer key

1. A% = (4.06 − 4.00)/4.00 × 100 = **+1.50%** (heavy). New draft = 7.6 × (4.06/4.00) = **7.71** — draft rises to bring weight down.
2. D = 110/π = **35 mm**. Expect an eccentric or damaged roller of 35 mm diameter (or its cot) in the drawframe that produced the sliver.
3. The can contains sliver made before the correction; testing it would judge the old setting. Only material produced after the change verifies the new draft — and a mixed can would also carry the fault into the next process unnoticed.

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